

19/11/25

Team - ID - 47

Students: Juan, Alvarez, Prathul Ravuri,
Prathav GuptaML - Exercise - 05Qn-2) Classification and Regression Trees (CART): Hands-On1) $N=5$, $K=3$, $V = \{\text{Tennis, Football, Basketball}\}$, $P(V) = \{\frac{2}{5}, \frac{2}{5}, \frac{1}{5}\}$

(i) Best initial split

$$N \cdot H(V) = 5 \cdot [-0.4 \cdot \log_2 0.4 - 0.4 \log_2 0.4 - 0.2 \log_2 0.2]$$

$$= 5 \cdot [1.52] = 7.6$$

Possible Splits: [Let Ball Color be X_1 and Ball Width be X_2]

① Ball Color: $\rightarrow$ (i) $L = \{X_1^{\text{Yellow}}\}$, $R = \{X_1^{\text{White, Brown}}\}$ $\rightarrow P(V^L) = \{1, 0, 0\}$, $N^L = 2$
 $P(V^R) = \{0, \frac{2}{3}, \frac{1}{3}\}$, $N^R = 3$

(ii) $L = \{X_1^{\text{White}}\}$, $R = \{X_1^{\text{Yellow, Brown}}\}$ $\rightarrow P(V^L) = \{0, 1, 0\}$, $N^L = 1$
 $P(V^R) = \{\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\}$, $N^R = 4$

(iii) $L = \{X_1^{\text{Brown}}\}$, $R = \{X_1^{\text{Yellow, White}}\}$ $\rightarrow P(V^L) = \{0, \frac{1}{2}, \frac{1}{2}\}$, $N^L = 2$
 $P(V^R) = \{\frac{2}{3}, \frac{1}{3}, 0\}$, $N^R = 3$

Impurity Gains: $[N \cdot H(V) - N^L \cdot H(V^L) - N^R \cdot H(V^R)]$

I. i) $I = 7.6 - [2 \times 0 + 3 \times (-0.67 \cdot \log 0.67 - 0.33 \cdot \log 0.33)]$
 $= 7.6 - [0 + 2.74] = 4.86$

I. ii) $I = 7.6 - [1 \times 0 + 4 \times (-0.5 \log 0.5 - 2 \times 0.25 \log 0.25)]$
 $= 7.6 - [0 + 4 \times 0.5] = 1.6$

I. iii) $I = 7.6 - [2 \times (-2 \times 0.5 \times \log 0.5) + 3 \times (-0.67 \times \log 0.67 - 0.33 \times \log 0.33)]$
 $= 7.6 - [2 \times 1 + 2.74] = 2.86$

II) Ball Width: $[X_2 = \{6, 22\}]$ $\rightarrow$ Only midpoint is $v'' = 14$ $\rightarrow N^L = 2$, $P^L = \{1, 0, 0\}$
 $N^R = 3$, $P^R = \{0, \frac{2}{3}, \frac{1}{3}\}$

i) $I(X_2 \leq v'' = 14) = 7.6 - [2 \times 0 + 3 \times (-0.67 \times \log 0.67 - 0.33 \times \log 0.33)]$
 $= 4.86$

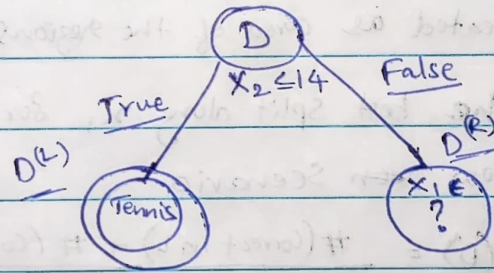
∴ The initial split that gives highest information gain is
 either Ball Color = {Yellow} & {not Yellow} or (Ball width ≤ 14)

1.2 Construction of entire ~~sub~~ decision tree

⇒ Let's consider the Ball width split as the first one.

⇒ The left subtree doesn't require any further split and can be assigned to Tennis.

⇒ The right subtree would need a split on Ball Color.

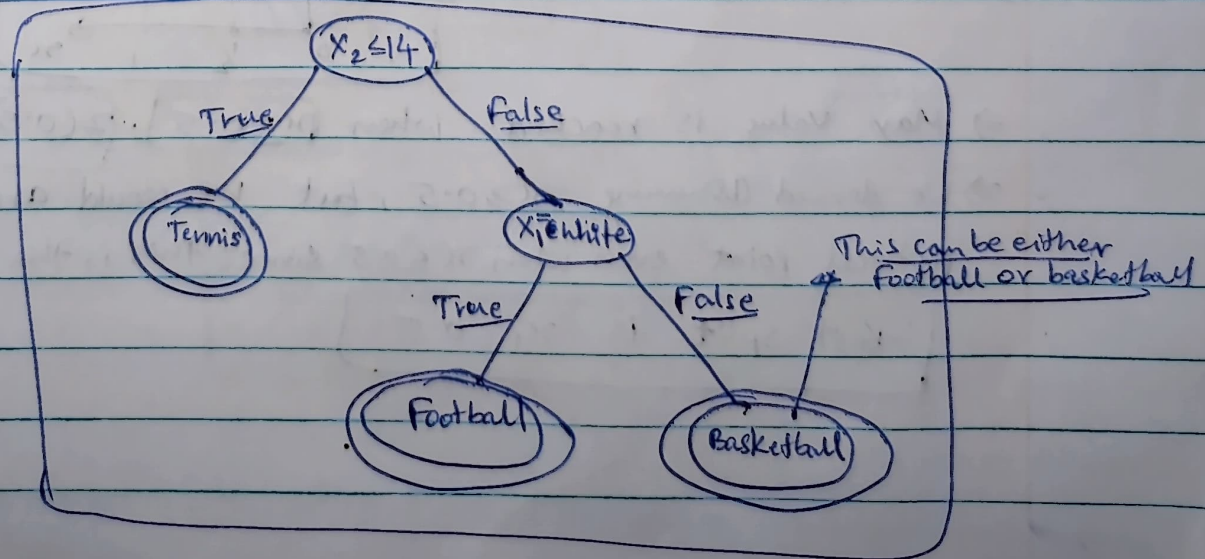


⇒ Let's calculate the information gain with the split $\begin{cases} L \in \text{White} \\ R \in \text{Brown} \end{cases}$.

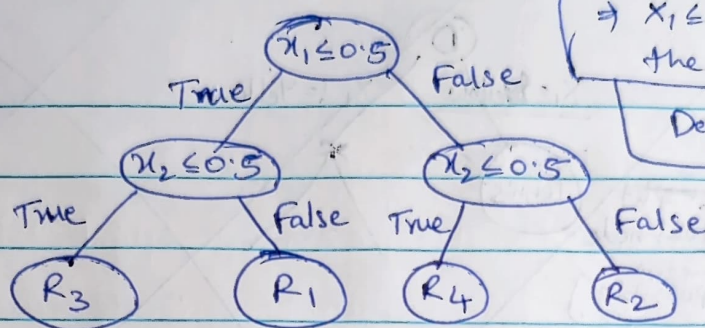
[Ball-width can't be split further, since it's $22 \neq d \in D(R)$.]

$$\begin{aligned} \Rightarrow \text{Info Gain} &= 3 \times (-0.67 \times \log 0.67 - 0.33 \times \log 0.33) - [1 \times 0 + 2 \times (-2 \times 0.5 \times \log 0.5)] \\ &= 2.74 - 2 = \boxed{0.74} \end{aligned}$$

The final Decision Tree



2.1) Decision Tree for the given region splits 2



$\Rightarrow x_1 \leq 0.5 \wedge x_2 \leq 0.5$ are the decision splits.

Depth of Tree = 2

2.2) $\Rightarrow$ A metric to match the requirements where 1 is the best metric accuracy score - and 0 is the worst can be ~~something derived from Gini Index~~

It can also be the accuracy score

$$\text{Metric} \equiv \frac{\#(\text{Correct Predictions})}{\#(\text{Total Predictions})}$$

[Let's refer to this Metric as $Z(D)$]

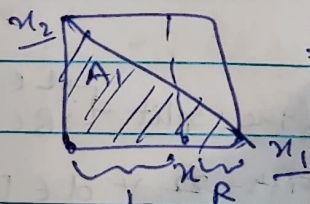
$\Rightarrow$ For the above decision tree it is as below -

$$Z(D) = \frac{1}{4} [Z(R_1) + Z(R_2) + Z(R_3) + Z(R_4)] = \frac{1}{4} \left[\frac{1}{2} + 1 + 0 + \frac{1}{2} \right] = 0.5$$

[Here number is treated as area of the regions, since it's continuous space]

2.3) Let's figure out the best split along x_1 for decision stump (depth=1).

$\Rightarrow$ Consider the below seen scenario -

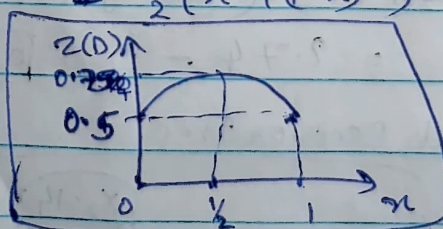


$$\Rightarrow Z(D) = \frac{\#(\text{Correct in L}) + \#(\text{Correct preds in R})}{\#(L+R)}$$

$$= \frac{(x \cdot \frac{1}{2}x^2) + ((1-x) \cdot \frac{1}{2}(1-x)^2)}{1} \quad [\text{Assuming } x \geq 0.5]$$

[Left would predict class-1 and R would predict class-0 (Assuming $x \geq 0.5$)]

$$= 1 - \frac{1}{2}(x^2 + (1-x)^2) = -x^2 + x + 0.5$$



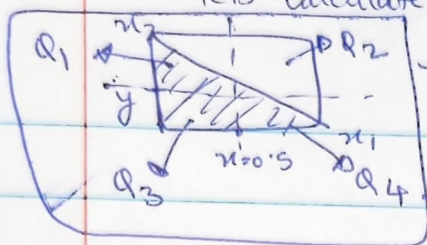
$\Rightarrow$ Max value is reached when $x \leq 0.5$ $\cdot$ $Z(0.5) \approx 0.75$

$\Rightarrow$ We derived assuming $x \geq 0.5$, but we would arrive at the same optimal point even when $x \leq 0.5$ since this is the limiting case.

$\therefore$ Best split is $x_1 \leq 0.5$

2.4) Next split (on axis x_2): Given $x_1 \leq 0.5$ as the first split,

→ let's calculate the best split of y on axis x_2 .



→ let calculate first for $y \leq 0.5$.

→ Let's call the regions as Q_1, Q_2, Q_3, Q_4 .

(Resultant)
 ~~$\Rightarrow Z(D) = \#(\text{Correct in } Q_1) + \#(\text{Correct in } Q_2) + \#(\text{Correct in } Q_3) + \#(\text{Correct in } Q_4)$~~
 ~~$\#(Q_1 + Q_2 + Q_3 + Q_4)$~~

$\Rightarrow Z(D) = \#(\text{Correct in } Q_1) + \#(\text{Correct in } Q_2) + \#(\text{Correct in } Q_3) + \#(\text{Correct in } Q_4)$

[Note that; Q_1 predicts class 1, Q_2 predicts class 0, Q_3 predicts class -1, Q_4 predicts class -1]

$$\Rightarrow Z(D) = \underbrace{\frac{1}{2} \left(\frac{1}{2}((1-y) + (0.5-y)) \right)}_{Q_1} + \underbrace{\left(\frac{1}{2}(1-y) - \frac{1}{2}(0.5-y)^2 \right)}_{Q_2} + \underbrace{(0.5y)}_{Q_3} + \underbrace{\left(\frac{1}{2}(y) - \frac{1}{2}(y)(y) \right)}_{Q_4}$$

$$\Rightarrow Z(D) = \left(\frac{3}{8} - \frac{y}{2} \right) + \left(\frac{1}{2} - \frac{y}{2} - \left(\frac{y^2}{2} - \frac{y}{2} + \frac{1}{8} \right) \right) + \left(\frac{y}{2} \right) + \left(\frac{y}{2} - \frac{y^2}{2} \right)$$

$$Z(D) = -y^2 + y/2 + 0.75$$

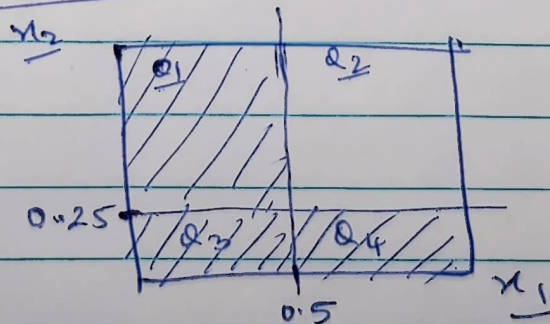
$\Rightarrow \text{Max}^m$ occurs at $y = 0.25$ and $z(0.25) = -\frac{1}{16} + \frac{1}{8} + \frac{6}{8} = \frac{13}{16}$

$\Rightarrow$ Due to symmetry a split at $y = 0.75$ gives the same z -score.

$\therefore$ Next best split would be $x_1 \leq 0.25$ & $x_2 \leq 0.75$

Resultant Decision Tree Regions

Option-1



Option-2

